

20. $\lim_{t \rightarrow 1} \frac{t^4 - 1}{t^3 - 1}$

14. $\lim_{x \rightarrow 4} \frac{x^2 + 3x}{x^2 - x - 12}$

DAY 6: CONTINUITY

Lecture Goals

- Understand the idea of the continuity intuitively
- Learn the precise (ϵ - δ) definition of the continuity
- Connect continuity w/ limits & function values
- Identify different types of discontinuities
- Practice determining where functions are continuous

THE BIG PICTURE

Calculus is built on the idea of small changes. If as x changes a little $f(x)$ only changes a little, we say the function is continuous. Continuity tells us there are no sudden jumps, breaks, or holes in the graph.

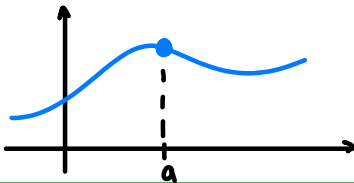
WHAT DOES IT MEAN FOR f TO BE CONTINUOUS?

A function f is a continuous @ a number a if the value of $f(x)$ gets arbitrarily close to $f(a)$ as x gets arbitrarily close to a .

↳ **THINK OF IT THIS WAY**: You are standing @ $x=a$. You can zoom in as close as you want. If the graph of f has no breaks, jumps, or holes @ that point, then f is continuous @ a .

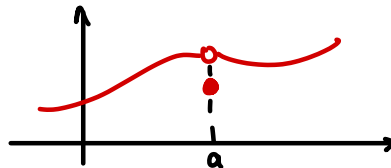
CONTINUOUS

You can draw the graph w/out lifting your pencil.



NOT CONTINUOUS

The graph has a break, a jump, or hole @ $x=a$.



DEFINITION: CONTINUITY

Definition (continuity @ a point)

A function f is **continuous** at a number a in its domain if the following 3 conditions are satisfied:

① $f(a)$ is defined

② $\lim_{x \rightarrow a} f(x)$ exists

③ $\lim_{x \rightarrow a} f(x) = f(a)$

IN WORDS:

The function is defined at the limit exists, and the limit exists equal the function value.

NOTE

↳ all conditions must be TRUE

↳ If even ONE fails, f is not continuous @ a .

Continuity is about the behavior of the function near the point. NOT just AT the point.

Examples

say a function has the following limits & value @ a

#1 $f(3) = 10$

$$\lim_{x \rightarrow 3} f(x) = 10$$

yes $\lim_{x \rightarrow 3} f(x) = f(3) \checkmark$

#2 $f(2) = \text{DNE}$

$$\lim_{x \rightarrow 2} f(x) = 4$$

No $\lim_{x \rightarrow 2} f(x) \neq f(2)$

#3 $f(5) = 2$

$$\lim_{x \rightarrow 5} f(x) = 4$$

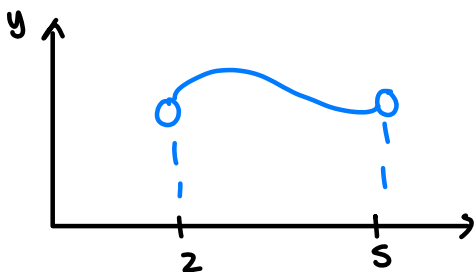
No $\lim_{x \rightarrow 5} f(x) \neq f(5)$

CONTINUITY ON INTERVAL

OPEN INTERVALS ()

Definition A function f is continuous on an open interval (a,b) if it is continuous @ every number c in (a,b) .

- Only check interior points
- DO NOT CHECK END POINTS



Example: on $(2,5)$
continuity on $(2,5)$
only asks for
points between them
endpoints don't count

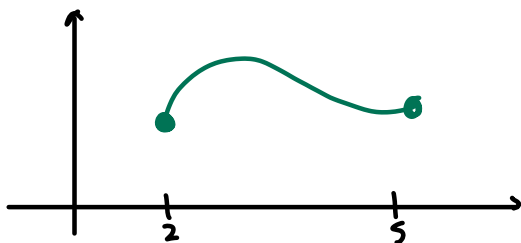
CLOSED INTERVALS []

A function f is continuous on a closed interval $[a,b]$

- ↳ if it is continuous on (a,b)
- ↳ continuous from **right** @ a
- ↳ continuous from **left** @ b

⇒ AT ENDPOINTS ⇐

- f is continuous from the right @ a if: $\lim_{x \rightarrow a^+} f(x) = f(a)$
- f is continuous from left at b if: $\lim_{x \rightarrow a^-} f(x) = f(b)$



$$\lim_{x \rightarrow 2^+} f(x) = f(2)$$

$$\lim_{x \rightarrow 5^-} f(x) = f(5)$$

HALF INTERVALS

↳ one endpoint is included, the other is NOT

• in these cases:

1) check continuity on $(a|b)$

2) For the closed interval part

↙
 $[a|b)$

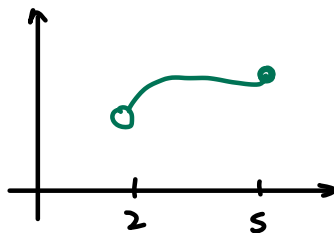
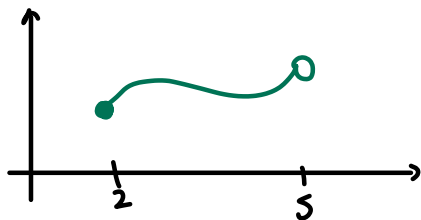
check $\lim_{x \rightarrow a^+} f(x)$

and make sure $= f(a)$

↘
 $(a|b]$

check $\lim_{x \rightarrow b^-} f(x)$

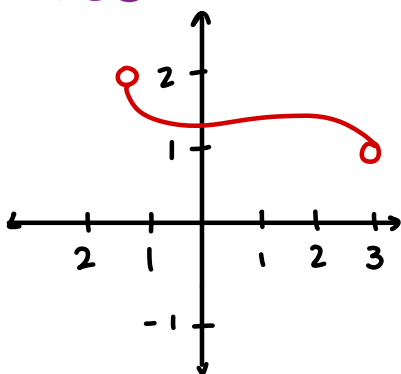
and make sure $= f(b)$



HOW TO READ FROM GRAPHS

- ① Identify the interval (open, closed, or half open)
- ② check all interior points for holes, jumps, breaks, or vertical asymptotes
- ③ Decide whether endpoints are included
- ④ For included endpoints, use one sided continuity:
 - ↳ RIGHT LIMIT @ the left endpoint (-)
 - ↳ LEFT LIMIT @ the right endpoint (+)

Examples

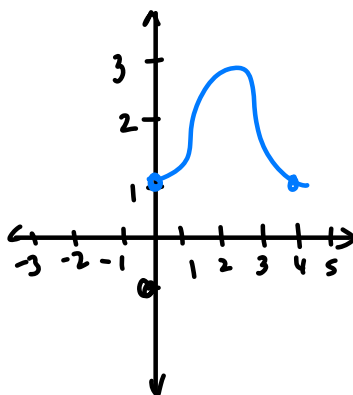


continuous on $(-1, 3)$

BUT NOT ON $[-1, 3]$

Why? Left endpoint has open circle &

So does Right endpoint

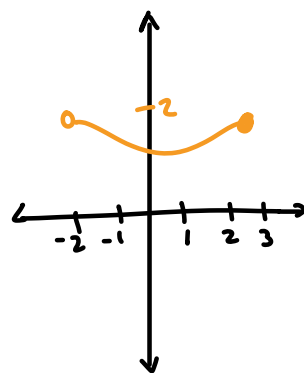


continuous on $[0, 4]$
smooth inside and endpoints

match

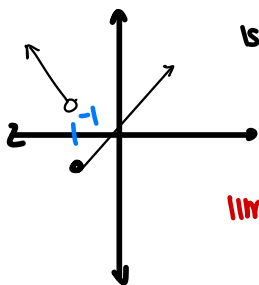
$$\lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\lim_{x \rightarrow 4^-} f(x) = f(4)$$



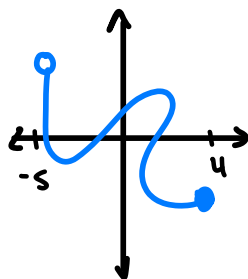
continuous on $(-2, 2]$ but
Not on $[-2, 2]$
left endpoint

YOU try:



Is this continuous?

No
At -1
lim does not exist



Is this continuous

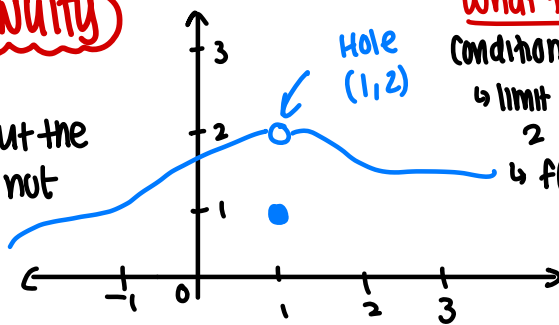
$[-5, 4]$?

No! but
 $(-5, 4)$ yes

COMMON DISCONTINUITIES

REMOVEABLE DISCONTINUITY (HOLE)

The limit exists and is finite, but the function value is missing or not defined correctly



What fails?

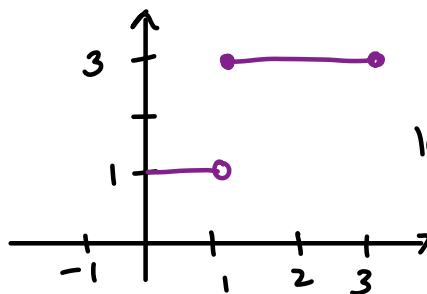
(Condition 1) or 3)

↳ limit exists = 2

↳ $f(1) \neq 2$

JUMP DISCONTINUITY

left and right limit exists (finite) but they are different



What fails?

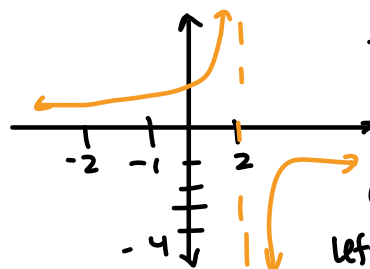
condition 2) & 3)
left limit $\neq$ right

so

$$\lim_{x \rightarrow 1} f(x) = DNE$$

INFINITE DISCONTINUITY

At least one of the one-sided limits is $+\infty$ or $-\infty$ Vertical asymptote



What fails?

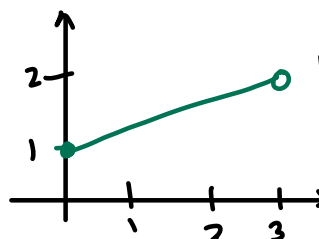
The limit is not finite

so it does not equal to real #

Left = ∞ Right = $-\infty$

ENDPOINT / OPEN CIRCLE

At an endpoint the function may not be included. Then the function is not continuous on a closed interval.



What fails?

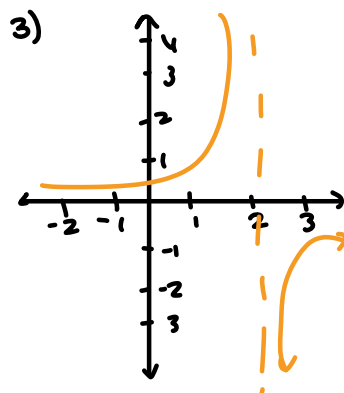
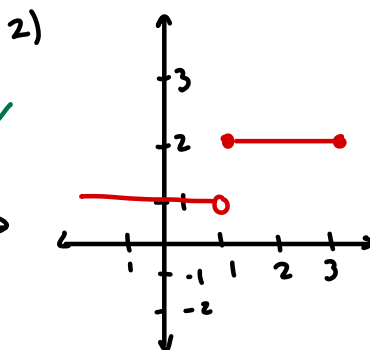
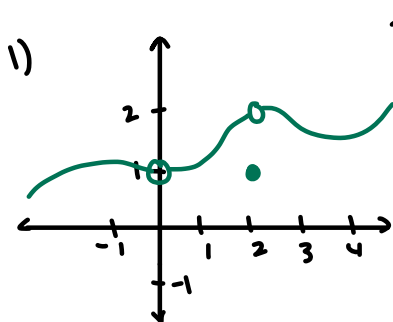
1) & 3)

$x=3$ not defined

so not cont. on $[0, 3]$

is cont. on $[0, 3)$

You TRY? Identify type of discont.



Continuity Algebraically

ALGEBRAIC STRATEGY

- STEP 1 Find the point a
- STEP 2 Check whether $f(a)$ is defined
- STEP 3 Find $\lim_{x \rightarrow a} f(x)$
- STEP 4 Compare the limit to $f(a)$
- STEPS State whether the function is continuous & if not, what type of discontinuity.

COMMON TRAPS

- denominator = 0
- forgetting point must be in domain
- mixing up limit & function values

Example

Ex 1: If $f(x) = x^2 + 3x - 4$ cont @ $x = 2$?

- 1) Find point $a = 2$
- 2) check $f(2) = 2^2 + 3(2) - 4 = 6$
- 3) find limit.
 $\lim_{x \rightarrow 2} x^2 + 3x - 4 = 6$
- 4) $\lim_{x \rightarrow 2} f(x) = 6$ $f(2) = 6$
- 5) Yes continuous.

polynomials tend to be continuous

Ex 3: Is $h(x) = \frac{1}{x-3}$ continuous at $x = 3$

- 1) $a = 3$
- 2) $h(3) = \frac{1}{3-3} = \frac{1}{0} = \text{und}$
- 3) Find limit
 $\lim_{x \rightarrow 3^+} \frac{1}{x-3} = -\infty$ $\lim_{x \rightarrow 3^-} \frac{1}{x-3} = \infty$ vertical asympt.

- 4) compare limit DNE & $h(3)$ undefined
- 5) Not continuous

Ex 2 Is $g(x) = \frac{x^2 - 1}{x - 1}$ continuous

@ $x = 1$?

- 1) $a = 1$
- 2) $g(1) = \frac{1^2 - 1}{1 - 1} = \frac{0}{0} = \text{und}$
- 3) limit:
 $\frac{x^2 - 1}{x - 1} = \frac{(x+1)(x-1)}{\cancel{x-1}} = x+1$
 $\lim_{x \rightarrow 1} x+1 = 1+1 = 2$
- 4) limit = 2 but $g(1) = \text{und}$
- 5) Not continuous

HOW TO STATE INTERVALS OF CONT.

- State all x-values when the function is continuous
- Break the real line @ pt where the function is not defined
- Use open intervals between breaks
- Include endpoints only if the function is defined & continuous there

YOU TRY

a) $p(x) = 5x^2 - 1$ @ $x=0$

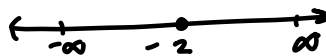
- $p(0) = 5(0) - 1 = -1$ (defined)
- $\lim_{x \rightarrow 0} 5x^2 - 1 = 5(0)^2 - 1 = -1$
- $\text{limit} = -1 \neq p(0) = -1$
cont. ✓

b) $q(x) = \frac{x+2}{x+4}$ @ $x=4$ Not cont @ $x=4$

ex: 

Not defined @ $x=3$ Cont.
 $(-\infty, 3) \cup (3, \infty)$

ex: $g(x) = \sqrt{x+2}$



domain $x \geq -2$
cont. on $[-2, \infty)$

ex: $p(x) = x^3 - 4x +$



Poly are cont. anywhere

PIECEWISE FUNCTIONS

A piecewise function uses different formulas on different intervals of x .

$$f(x) = \begin{cases} x+1 & \text{if } x < 0 \\ 2 & \text{if } x = 0 \\ x^2 & \text{if } x > 0 \end{cases}$$

example:

$$\text{if } f(x) = \begin{cases} x+1 & \text{if } x < 1 \\ 3-x & \text{if } x \geq 1 \end{cases} \quad \begin{aligned} f(0) &= 0+1 = 1 \\ f(2) &= 3-2 = 1 \\ f(1) &= 3-1 = 2 \end{aligned}$$

CONTINUITY FOR PIECEWISE

Piecewise functions can only be discontinuous when the formula changes - @ the join points (boundary values)

At each join point c check:

- left hand limit from left
- right hand limit from right
- Function value @ c

Ex:

$$\text{let } h(x) = \begin{cases} x+1 & \text{if } x < 1 \\ 3-x & \text{if } x \geq 1 \end{cases}$$

check continuity @ $x=1$

1) left hand limit:

$$\lim_{x \rightarrow 1^-} h(x) = \lim_{x \rightarrow 1^-} x+1 = 2$$

3) Function value

$$h(1) = 3-1 = 2$$

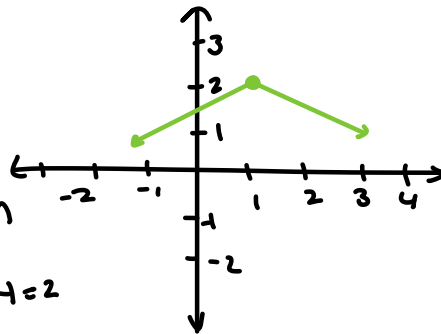
2) Right hand limit

$$\lim_{x \rightarrow 1^+} = \lim_{x \rightarrow 1^+} (3-x) = 2$$

4) Right = 2

Left = 2

$$h(1) = 2$$



You Try

1) let

$$f(x) = \begin{cases} 2x+1 & \text{if } x < 2 \\ 5 & \text{if } x \geq 2 \end{cases}$$

@ $x=2$

1) left hand limit

$$\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} (x+2) = 2$$

$$3) g(0) = 0^2 + 1 = 1$$

4) left = 2

right = 1

Not equal

2) Right hand limit

$$\lim_{x \rightarrow 0^+} g(x) = \lim_{x \rightarrow 0^+} (x^2 + 1) = 1$$

2) let

$$k(x) = \begin{cases} x^2 - 1 & \text{if } x < 0 \\ -x + 1 & \text{if } x \geq 0 \end{cases}$$

$x=0$